UNIVERSITA DI PISA

MASTER'S THESIS

SUPERVISED BY: PROFESSOR ALESSIO GIORGETTI

A Heuristic Approach to Routing and Spectrum Assignment in Multi-Fiber Elastic Optical Networks: Design and Implementation in TeraFlowSDN

By: ASM Chafiullah Bhuiyan

Computer Science and Networking

Academic Year: 2025–2026

Acknowledgement

First and foremost, I would like to express my deepest gratitude to my primary supervisor, Professor Alessio Giorgetti. I am incredibly thankful for his unwavering support, his willingness to share his resources, and for granting me welcoming access to his office. His generous investment of valuable time and expertise was instrumental in allowing me to complete this extensive and complex project within the given timeframe. Furthermore, I am deeply appreciative of his provision of dedicated hardware resources, which afforded me endless opportunities to conduct rigorous testing and successfully validate my work.

Finally, words cannot adequately express my profound gratitude to my family for their immense sacrifices during my time studying abroad. I owe a special and heartfelt thank you to my children, who have had to navigate a significant part of their childhood missing their father. Your love, patience, understanding, and resilience have been my greatest source of strength and motivation throughout this journey. This thesis is as much yours as it is mine.

Abstract

Thesis Title
Master's in Computer Science and Networking

The provisioning of light-paths in flexible-grid optical networks has significantly enhanced spectrum utilization; however, the integration of topologies featuring multiple parallel links between optical devices introduces profound computational complexities in Routing and Spectrum Allocation (RSA). Specifically, when multiple transponders (muxponders) interface with the exact same degree of Reconfigurable Optical Add-Drop Multiplexer (ROADM) devices, the permutation of viable routing options increases exponentially. This geometric expansion renders standard path computation mechanisms excessively heavy and inefficient for traditional Software-Defined Networking (SDN) controllers. This routing challenge is further compounded by the stringent constraints of spectrum continuity and contiguity under the ITU-T G.694.1 standard, which dictates a 12.5 GHz minimum slot width and a high-resolution 6.25 GHz slot granularity. Because optical signals directed to the same ROADM degree cannot reuse identical frequencies without severe interference, the controller is burdened with the computationally intensive task of continuously verifying the availability of every individual 6.25 GHz slot across all potential parallel routes. To overcome these dual bottlenecks of path scalability and high-resolution spectrum mapping, this thesis introduces the ParallelOpticalController, a novel microservice engineered directly into the `TeraFlowSDN` ecosystem. Leveraging the `networkX` library for advanced topological graph modeling, the proposed architecture natively supports multi-directed graphs with parallel edges. This enables the highly efficient execution of Dijkstra's algorithm to compute the optimal shortest path, while concurrently discovering and suggesting exhaustive alternative path combinations to maximize network utilization. Furthermore, to resolve the heavy spectrum calculation overhead, the controller employs an innovative bitwise mapping algorithm that performs mathematical intersections of available flex-grid slots across sequential hops, drastically reducing the time required to identify contiguous spectrum blocks. Finally, these backend computational advancements are fully integrated with a dynamic graphical WebUI that visualizes the complex parallel-link topology, endpoint interfaces, and real-time spectrum allocations, thereby offering a highly scalable, end-to-end framework for next-generation optical network management.

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1 Introduction

1.1 Overview

The unprecedented surge in global data traffic, driven largely by the widespread deployment of 5G cellular networks and Fiber-to-the-Home (FTTH) access architectures, has fundamentally reshaped the telecommunications landscape. To satisfy this monumental and continuous demand without incurring the prohibitive costs of laying new physical fiber, network operators are compelled to maximize the transmission capacity of existing optical resources. This is primarily achieved by aggregating multiple communication channels into a single optical fiber using Wavelength Division Multiplexing (WDM) and, more specifically, Dense Wavelength Division Multiplexing (DWDM), which increases fiber efficiency by transmitting independent signals at different wavelengths. However, as individual fiber capacity nears its theoretical

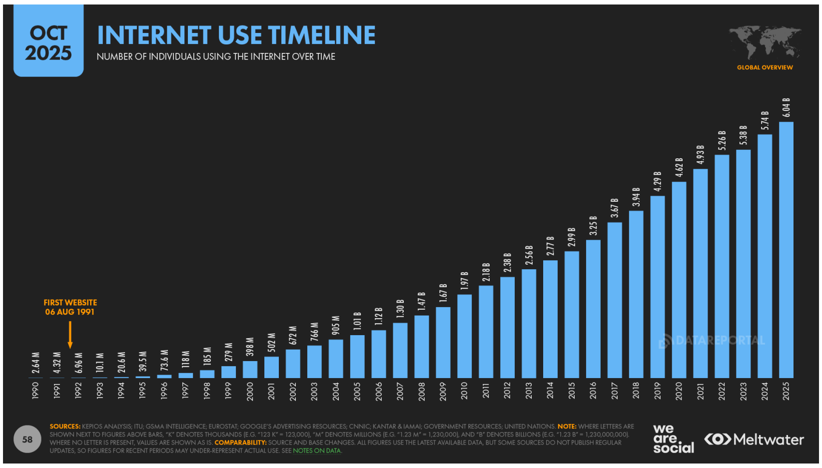

Figure 1.1: Number of internet users over the time

Shannon limits, spectral multiplexing alone is often insufficient to satisfy massive localized traffic requirements. Consequently, network operators increasingly deploy "multifiber" or parallel-link configurations—utilizing multiple physical fiber pairs between adjacent nodes—to achieve a multiplicative increase in total throughput. While protocols like Link Aggregation (LAG) are traditionally utilized at higher network layers to duplex or group these connections for redundancy and increased bandwidth, managing these resources at the optical layer introduces significant complexity.

To handle the management burden of these high-capacity, multifiber infrastructures, the optical control plane has transitioned from static, manual configurations to highly automated, centralized management systems empowered by Software-Defined Networking (SDN). As DWDM technologies have matured, the traditional fixed-grid spectrum allocation—which assigns rigid, uniform channel spacing—has become a structural bottleneck. To overcome this, flexible-grid (flex-grid) architectures have emerged as the industry standard. By adhering to ITU-T G.694.1 standards, which utilize a precise 6.25 GHz slot granularity with 12.5 GHz slot width, flex-grid networks allow for elastic, demand-driven bandwidth allocation that vastly improves overall spectral efficiency [1]. The deployment of flex-grid systems in long-haul backbone

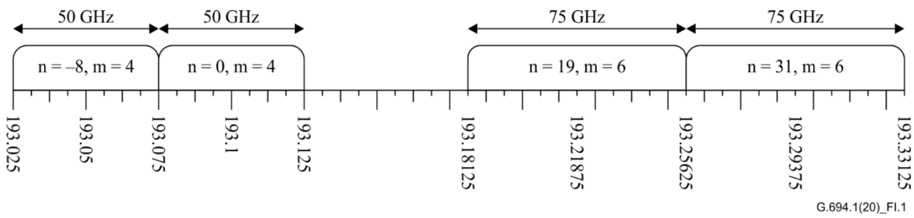

Figure 1.2: An example of use of flexible the grid

communications introduces substantial algorithmic complexity. In multi-hop networks, an optical signal must maintain both spectrum continuity—utilizing the same frequency across all links—and spectrum contiguity—utilizing adjacent frequency slots—from source to destination. This Routing and Spectrum Allocation (RSA) challenge is critically exacerbated in topologies featuring multiple parallel optical links between network elements, such as adjacent Reconfigurable Optical Add-Drop Multiplexers (ROADMs). When parallel links are introduced to a multi-hop architecture, the number of viable routing permutations increases exponentially. Consequently, traditional SDN controllers struggle to efficiently evaluate the availability of every 6.25 GHz slot across all possible path combinations, leading to severe computational bottlenecks and delayed provisioning times.

To resolve this critical scalability issue, this thesis proposes a novel approach by introducing the Parallel Optical Controller, a stateful microservice integrated directly into the TeraFlowSDN ecosystem. By leveraging advanced graph modeling techniques and a highly optimized bitwise spectrum intersection algorithm, this proposed method efficiently isolates available spectrum blocks across complex parallel routes. Ultimately, this solution effectively neutralizes the parallel link bottleneck, ensuring rapid, reliable, and highly scalable lightpath provisioning for next-generation optical networks.

1.2 Motivation

While extensive research has been conducted on Routing and Spectrum Allocation (RSA) in Elastic Optical Networks (EONs)[2][3],particularly concerning complex multi-hop lightpath provisioning[4]—there remains a notable gap in the literature regarding the efficient management of multiple parallel optical links between adjacent devices. In traditional centralized control planes, such as those managed by other SDNs, parallel links are typically treated as isolated, individual entities rather than aggregated resources[5]. Consequently, when a lightpath request is processed, the RSA algorithm is forced to evaluate the spectrum continuity of each parallel link independently[3]. This legacy approach is highly inefficient, as it triggers redundant, computationally heavy, back-and-forth calculations that severely degrade the controller's performance and scalability in dense optical topologies. The primary motivation of this thesis is to fundamentally resolve this parallel link computational bottleneck within the emerging TeraFlowSDN ecosystem [6]. By addressing relative algorithmic inefficiency, the research aims to design a system capable of evaluating multiple parallel paths concurrently, significantly reducing the computational overhead required to find available flex-grid spectrum.

Furthermore, theoretical RSA optimization cannot be solved just as a math problem or a simulation using static data; its efficacy is strictly dependent on the acquisition of precise, real-time telemetry from the underlying physical layer. While modern optical networks are rapidly transitioning toward vendor-agnostic architectures unified by standardized data models such as OpenConfig[7] and OpenROADM[8], a significant barrier remains in the practical deployment of centralized algorithms: hardware heterogeneity. Although these industry standards define common data models using YANG, individual equipment vendors frequently implement them with subtle variations in configuration hierarchies and telemetry reporting. Consequently, extracting uniform network state information—such as operational frequency ranges and transport capabilities—via a singular filtering procedure is often unfeasible in multi-vendor environments. This research addresses this disparity by bridging the gap between abstract path computation and physical hardware limitations through the extensive enhancement of device drivers within the TeraFlowSDN ecosystem. By refining the default driver logic and implementing a tiered approach that utilizes both standardized procedures and specialized, device-based handlers, this thesis establishes a more robust framework for multi-vendor support. These enhancements ensure that the proposed RSA algorithm is fed by accurate, model-driven telemetric data, effectively reconciling high-level computational logic with the practical complexities of diverse optical hardware management.

1.3 Research Objectives

To systematically address the challenges of Routing and Spectrum Allocation (RSA) over parallel optical links, this thesis is guided by the following core research objectives:

- **Analysis of Flexible-Grid Standards:** To investigate the operational principles of flexible-grid optical networks, specifically adhering to the ITU-T G.694.1 standards regarding spectrum slot width and granularity.

- **Evaluation of SDN Optical Intent Provisioning:** To study how optical intents and lightpath requests are currently handled by popular, industry-standard Software-Defined Networking (SDN) controllers, identifying their limitations in dense, multi-link topologies.

- **Examination of TeraFlowSDN Device Drivers:** To conduct a comprehensive study of the telemetry and configuration mechanisms within TeraFlowSDN, with a specific focus on analyzing the OpenConfig and OpenROADM device drivers.

- **Assessment of the Existing Optical Controller:** To critically evaluate the official OpticalController microservice currently operating within the TeraFlowSDN ecosystem, identifying its architectural constraints when processing complex optical intents.

- **Review of RSA Algorithmic Approaches:** To explore and analyze contemporary algorithmic strategies used to solve the RSA problem in multi-hop, multi-fiber optical networks, evaluating their computational efficiency and spectrum assignment mechanisms.

- **Formulation of an Optimal Solution:** To synthesize the findings from the aforementioned studies and propose an optimal, highly scalable RSA framework that efficiently resolves the computational bottlenecks associated with parallel optical links.

1.4 Thesis Contributions

Building upon the established research objectives, this thesis makes several tangible contributions to the field of optical Software-Defined Networking (SDN), specifically addressing the computational bottlenecks associated with routing and spectrum assignment over parallel links. The core contributions of this work are summarized as follows:

- **Enhancement of Optical Device Drivers:** A significant extension of the existing TeraFlowSDN device drivers, specifically the OCDriver (for OpenConfig devices) and the OpenROADM driver. This contribution includes the implementation of custom filters to extract vendor-agnostic telemetry—such as minimum and maximum operating frequencies, transport types (e.g., OCH, OMS), and DWDM parameters—directly from XML configurations, seamlessly integrating this data into an extended Context database schema.

- **Design of an Optimized RSA Algorithm:** The formulation and implementation of a highly scalable Routing and Spectrum Allocation (RSA) algorithm tailored for topologies with multiple parallel links. This includes utilizing the `networkX` library to construct multi-directed graphs that accurately represent parallel edges, alongside the introduction of a novel bitmap-based bitwise intersection method designed to rapidly compute the availability of contiguous spectrum slots across multiple hops respecting the parallel links.

- **Development of the ParallelOpticalController Microservice:** The primary contribution of this thesis is the design, implementation, and seamless integration of a novel, stateful microservice—the ParallelOpticalController—into the TeraFlowSDN ecosystem, which serves as the practical execution engine for the proposed Routing and Spectrum Allocation (RSA) algorithm. This implementation is explicitly engineered to navigate multi-hop topologies featuring parallel optical links between nodes at each hop, whereas traditional controllers often struggle with the exponential increase in viable routing permutations. The controller autonomously calculates paths that strictly adhere to both spectrum continuity and contiguity constraints, ensuring valid end-to-end lightpath establishment across disjoint physical resources while utilizing optimized logic to mitigate the computational bottlenecks associated with solving NP-hard routing problems in large-scale networks. By facilitating the high-speed discovery of primary shortest paths and evaluating alternative route combinations in real-time, the architecture provides robust support for dynamic traffic demands and enhances overall network resilience.

- **Advanced Network Visualization and WebUI Integration:** To comprehensively upgrade the TeraFlowSDN Web User Interface (WebUI) to reflect the complex physical and logical states of the optical network. This objective focuses on developing visual representations of device endpoint capabilities, real-time spectrum slot availability (based on ITU-T 6.25 GHz granularity), multi-hop topological links, and the dynamic output of the RSA computation.

By fulfilling these objectives, this research aims to deliver an end-to-end solution that spans from low-level physical device configuration up to high-level algorithmic path computation and graphical user interaction.

1.5 Organization of the Thesis

The remainder of this thesis is logically structured to guide the reader through the theoretical background, the proposed algorithmic solutions, the system implementation, and the final evaluation. The document is organized as follows:

- **Chapter 2:** Background and Literature Review provides the foundational concepts necessary to understand the research. It discusses the principles of Software-Defined Networking (SDN) in optical networks, introduces the TeraFlowSDN architecture, and details Optical Transport Networks (OTN) including WDM technology and ITU standards. The chapter concludes with a review of existing Routing and Spectrum Allocation (RSA) algorithms, highlighting the specific problems and limitations associated with parallel links.

- **Chapter 3:** Problem Definition and Proposed Algorithm formally establishes the research problem and introduces the mathematical formulation of the challenge. It details the graph modeling techniques used to represent topologies with parallel links and breaks down the proposed RSA algorithm, covering bandwidth computation, path discovery, bitmap-based spectrum allocation, and slot acquisition, followed by a thorough complexity analysis.

- **Chapter 4:** System Design and Implementation delves into the practical engineering and software architecture of the proposed solution. It outlines the modifications made to TeraFlowSDN—including device driver enhancements for OpenConfig and OpenROADM YANG models, database schema extensions, and the RSA Computation Module. Furthermore, it details the development of the new Parallel Optical Controller microservice and the corresponding WebUI enhancements for spectrum visualization.

- **Chapter 5:** Testbed Setup and Deployment outlines the experimental environment used to validate the proposed system. It describes the physical and virtual infrastructure, the configuration of emulated transponders and ROADM devices, the setup of network topologies, and the deployment procedure utilizing MicroK8s.

- **Chapter 6:** Results and Evaluation presents a comprehensive analysis of the system's performance. It covers functional validation, computational performance, and spectrum utilization efficiency. The chapter also provides a throughput analysis, a comparative study against baseline approaches, and a statistical discussion of the results.

- **Chapter 7:** Conclusion and Future Work summarizes the primary contributions of the thesis, discusses the current limitations of the implemented system, and suggests potential directions for future research and development.

2 Background and Literature Review

This chapter establishes the theoretical foundation and contextual background for the research presented in this thesis, tracing the technological progression from foundational Optical Transport Networks (OTN) to modern automated control architectures. The review begins by examining the evolution of Wavelength Division Multiplexing (WDM), detailing the industry-wide transition from rigid, fixed-grid spectrum allocation to the highly efficient, elastic paradigms of flexible-grid (flex-grid) EONs. It contrasts traditional, decentralized optical control planes with the emergence of Software-Defined Networking (SDN), which provides the centralized intelligence required for dynamic resource management. Within this context, the chapter details the architectural components of contemporary SDN controllers, specifically focusing on the TeraFlowSDN ecosystem. A critical analysis of current Routing and Spectrum Allocation (RSA) methodologies follows, identifying a persistent research gap: the inability of traditional path computation and spectrum assignment logic to scale effectively when encountering multi-hop topologies with parallel optical links. Finally, the chapter introduces the `ParallelOpticalController` as a transformative solution within TeraFlowSDN, designed to resolve these computational bottlenecks and provide high-speed, constraint-aware provisioning for complex multifiber infrastructures

2.1 Software Defined Network

2.1.1 SDN Architecture and Principles

Software-Defined Networking (SDN) represents a fundamental paradigm shift in telecommunications, decoupling the network's control logic—the control plane—from the underlying packet or optical forwarding hardware, known as the data plane. This structural separation centralizes network intelligence into a software-based controller that maintains a global, real-time view of the network topology and its available resources. By abstracting the physical infrastructure through open, standardized Southbound Interfaces (SBIs) such as NETCONF or gNMI, SDN enables deep network programmability and dynamic orchestration of forwarding behaviors. This centralized management framework allows high-level applications to request network resources through Northbound Interfaces (NBIs) without requiring granular knowledge of the physical layer's complexity. Consequently, SDN provides the necessary architecture to achieve multi-vendor interoperability and agile service provisioning, transforming rigid physical infrastructure into flexible, automated, and software-governed assets.

2.1.2 SDN in Optical Networks

While initially designed for packet-switched networks, SDN principles have been aggressively adapted for the optical domain to create Software-Defined Optical Networks (SDON). Unlike packet networks, optical networks must manage analog physical layer impairments, wavelength continuity, and complex spectrum allocation constraints. SDON abstracts these physical layer complexities by decoupling the intelligent control logic—termed Optics Defining Software (ODS)—from the programmable physical hardware, known as Software Defined Optics (SDO) [9]. This abstraction allows the central controller to dynamically provision lightpaths, tune variable transponders for specific modulation formats, and automatically configure advanced Reconfigurable Optical Add-Drop Multiplexers (ROADMs), successfully moving away from historically static, manually operated optical grids. To manage this complexity at scale, modern SDON deployments frequently adopt a Transport-SDN (T-SDN) architecture, this is often achieved through a hierarchical control structure [10]. In this model, lower-level controllers manage specific optical nodes, while a higher-level controller (such as ONOS) facilitates multi-vendor integration using standardized interfaces like the Transport API (T-API) over NETCONF and RESTCONF protocols. Furthermore, breaking vendor lock-in is a critical focus of modern optical control planes and initiatives like the Open Disaggregated Transport Network (ODTN) utilize standardized YANG data models (such as OpenConfig and OpenROADM) to abstract heterogeneous hardware [11]. This standardization enables the seamless, end-to-end integration of packet and optical layers across disaggregated white-box devices. Finally, as emphasized in "Network Slicing in SDN Networks," this centralized, programmable control is what ultimately enables advanced network virtualization. By leveraging SDON capabilities, operators can deploy multiple isolated logical slices over a single shared physical infrastructure, dynamically shaping traffic to guarantee specific Quality of Service (QoS) and performance isolation for varying applications.

2.2 TeraflowSDN

2.2.1 Architecture and Ecosystem

TeraFlowSDN (TFS) is a state-of-the-art, open-source SDN controller developed under the European Telecommunications Standards Institute (ETSI) to radically advance the orchestration of beyond 5G and 6G networks. Designed as a secure, cloud-native platform, TFS employs a microservices-based architecture orchestrated within Kubernetes environments, enabling individual control components to operate autonomously and scale independently. This inherent modularity provides an ideal ecosystem for integrating custom logic, such as vendor-agnostic device drivers and novel path computation engines. Furthermore, the architecture is explicitly engineered to seamlessly integrate with modern Network Function Virtualization (NFV) and Multi-access Edge Computing (MEC) frameworks, including ETSI Open-SourceMANO. By unifying network and cloud resource management, TeraFlowSDN equips telecom operators and hyperscale cloud providers with zero-touch automation, high business agility, and dynamic service provisioning across complex, multi-tenant optical and packet transport infrastructures.

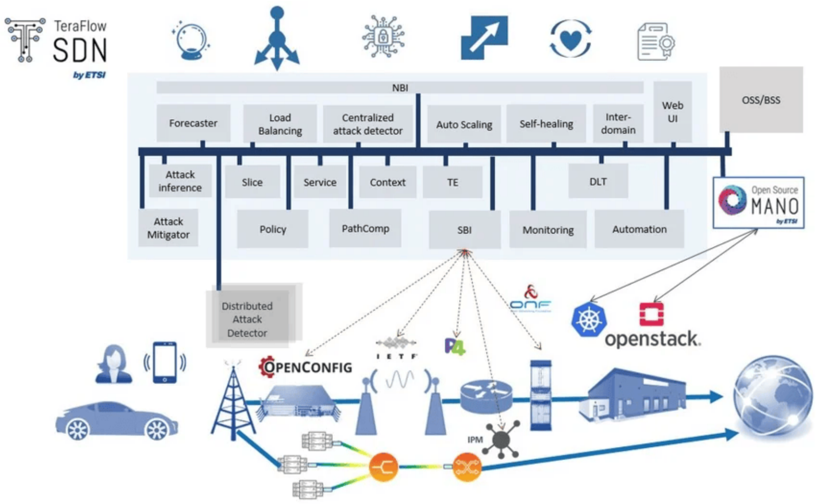

Figure 2.1: TeraflowSDN Architecture

2.2.2 Key services

The TeraFlowSDN ecosystem relies on a highly coordinated suite of specialized microservices, communicating via Remote Procedure Calls (gRPC), to maintain network state and execute dynamic configurations. At the core of this architecture is the Context Service, which acts as the highly available, stateful storage mechanism maintaining the definitive global view of multi-domain topologies, devices, endpoints, and active network slices. Interacting closely with this centralized database is the Device Service, which manages all southbound interfaces by utilizing a modular framework of vendor-agnostic drivers—often leveraging NETCONF and OpenConfig models—to dynamically onboard and configure physical or emulated hardware. Furthermore, the Service Component oversees the end-to-end lifecycle of network connections by interpreting incoming connectivity requests and delegating cross-layer configurations to appropriate handlers. For optical transport infrastructures specifically, the OpticalController operates as a dedicated service to process complex optical intents, executing resource allocation algorithms such as Routing and Spectrum Assignment (RSA) to establish continuous and contiguous lightpaths across multi-hop networks. These fundamental operations are visually managed through a graphical WebUI that provides topology visualization and telemetry display, while the broader ecosystem is fortified by complementary microservices dedicated to machine learning-based security, distributed ledger technologies for multi-tenancy, and policy-driven automation to ensure robust, zero-touch network orchestration.

2.3 Optical Transport Networks

2.3.1 WDM technology

Wavelength Division Multiplexing (WDM), specifically Dense WDM (DWDM), is the foundational technology for modern high-capacity optical networks. DWDM enables the transmission of multiple data streams over a single optical fiber by modulating each stream onto a distinct wavelength (or color) of laser light. Historically, DWDM utilized a fixed-grid spacing, allocating rigid channel intervals of 50 GHz, 100 GHz, or even wider. However, as data rates increased and traffic patterns became more unpredictable, this rigid spacing resulted in inefficient spectrum utilization. To address this bottleneck, modern elastic optical networks (EONs) utilize flexible-grid (flex-grid) technologies[12]. As defined in the ITU-T G.694.1 standard, the flex-grid allows a mixed bit rate or mixed modulation format transmission system to allocate frequency slots with different widths, optimizing the spectrum for the specific bandwidth requirements of individual channels[1]. This transition from fixed to flexible allocation is what necessitates advanced Routing and Spectrum Allocation (RSA) algorithms, such as the one proposed in this thesis, to dynamically manage the resulting variable-width spectrum blocks across complex, multi-fiber topologies.

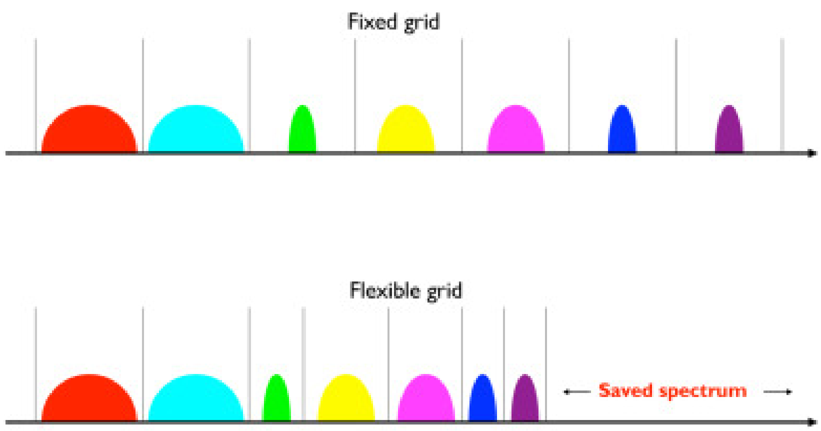

Figure 2.2: DWDM Flex-Grid Overview

2.3.2 ITU standards

The implementation of these flex-grid networks is strictly governed by International Telecommunication Union (ITU) standards to ensure interoperability and consistent spectrum management. Specifically, the ITU-T G.694.1 Recommendation provides the definitive frequency grid for DWDM applications. Under this standard, the entire DWDM spectrum is anchored to a reference central frequency of 193.1 THz. To enable elastic allocation, the standard defines a "frequency slot" as a contiguous frequency range characterized by its nominal central frequency and its total slot width. The architecture is built upon two critical granularities:

- **Nominal Central Frequency Granularity:** The allowed central frequencies are defined by the formula $193.1 + n \times 0.00625$ THz, establishing a fine-grained granularity of 6.25 GHz to allow precise positioning of channels.

- **Slot Width Granularity:** The total width of any assigned frequency slot is defined by $12.5 \times m$ GHz, establishing 12.5 GHz as the minimum, indivisible building block for bandwidth allocation.

2.3.3 Network elements (devices)

The physical topology comprises two primary categories of optical nodes: terminal multiplexers (transponders and muxponders) and Reconfigurable Optical Add-Drop Multiplexers (ROADMs). In modern high-capacity optical networks, network endpoints are more precisely characterized as muxponders, which aggregate multiple lower-rate client-layer signals into high-speed optical channels suitable for DWDM transmission. These devices utilize bidirectional ports equipped with advanced pluggable transceivers that handle complex modulation and Forward Error Correction (FEC). While the telecommunications industry employs a diverse array of transceiver form factors and optical modules—such as QSFP, QSFP28, OpenZR+, and 400ZR—each offering varying frequency band support, capacity, and error correction capabilities, this thesis adopts a specific, standardized hardware profile for its emulation environment. Specifically, the emulated muxponder ports are modeled using the QSFP56-DD form factor equipped with Digital Coherent Optics, utilizing a 400GBASE-ZR Ethernet Physical Medium Dependent (PMD) sublayer and automated FEC. Although real-world deployments may feature heterogeneous transceiver capabilities, a foundational assumption of this research is that all muxponder ports are fully bidirectional and provide continuous, tunable frequency support across the entire C-band. This baseline ensures a uniform spectrum allocation space for evaluating the proposed RSA algorithms. Conversely, ROADMs serve as the optical switching backbone, enabling individual wavelengths to be dynamically added, dropped, or passed through a node entirely in the optical domain, thereby bypassing costly optical-to-electrical-to-optical (O-E-O) conversions. To accurately evaluate the Routing and Spectrum Allocation (RSA) algorithm on parallel links, it is necessary to establish strict assumptions regarding the internal architecture of these ROADMs. A typical ROADM architecture consists of Shared Risk Groups (SRGs) for local add/drop functionality and Direction Degrees (DEGs) equipped with Wavelength Selective Switches (WSS) for line-side routing. In the context of this thesis, physical fiber connections are only classified as strict "parallel links" if the ports originating from the same muxponder terminate at the same SRG and are routed outward

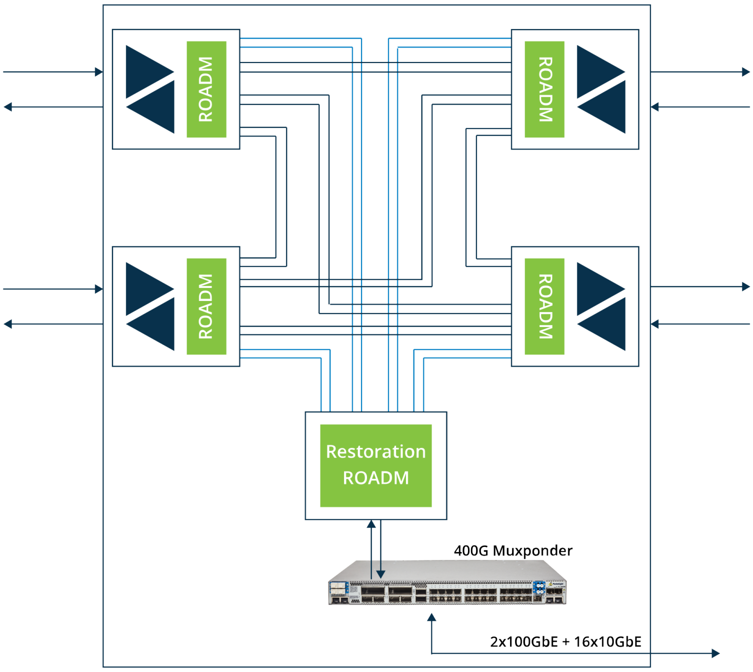

Figure 2.3: An example optical network

through the exact same DEG of the ROADM. Under this specific topological constraint, the parallel fibers share an identical physical routing vector, making spectrum contiguity and continuity algorithms highly complex, as the spatial routing does not inherently resolve spectrum overlap. If the ports were to route through different DEGs, the spatial separation provided by distinct WSS hardware would allow for simple frequency re-use, thereby negating the specific parallel-link RSA constraints addressed in this study. Furthermore, to accommodate high-capacity throughput and scalability, the ROADM devices in this simulated environment are assumed to be wideband-capable, operating seamlessly across the entire S, C, and L (SCL) optical bands.

2.3.4 Data models (configuration)

To achieve true interoperability and break the traditional hardware lock-in imposed by proprietary management interfaces, modern optical equipment relies on standardized YANG data models. OpenConfig[7] and OpenROADM[8] have emerged as the two prominent industry standards used to formalize the representation of optical device configurations and operational states. By structuring parameters in universally understood XML or JSON formats, these models hide the underlying hardware idiosyncrasies from the control plane, allowing an SDN controller to parse telemetry and push configurations uniformly regardless of the equipment manufacturer. To manage the complexity of optical networking, these data models utilize strictly defined XML namespaces to isolate specific operational domains. Within the emulation environment of this thesis, the terminal devices (muxponders) primarily utilize the

OpenConfig standard. For instance, the physical hardware hierarchy—such as form factors and transceivers—is defined under the plaform-namespace. Concurrently, the operational optics are managed via the terminal-device namespace, which dictates precise parameters such as operational frequencies (e.g., 191.6 THz), target output power, and the logical channel assignments mapping Ethernet client traffic to Optical Transport Network (OTN) line ports.

Conversely, the emulated optical switches utilize the OpenROADM data model, which is specifically tailored to describe complex photonic switching architectures. Using namespaces such as device namespace and interface namespace, the model accurately maps the internal ROADM topology. It defines critical switching components including circuit packs (such as twin Wavelength Selective Switches), Direction Degrees (DEGs), Shared Risk Groups (SRGs), and specific optical interfaces like Optical Transport Section (OTS) and Optical Multiplex Section (OMS). By adhering to these standardized schemas, the SDN controller is decoupled from vendor-specific command-line interfaces. Consequently, network operators can seamlessly deploy heterogeneous, multi-vendor optical topologies. A single controller can dynamically orchestrate an end-to-end lightpath by pushing standard OpenConfig XML payloads to tune a Cisco transceiver, while simultaneously pushing OpenROADM payloads to configure cross-connects on an Ericsson ROADM, effectively unifying the entire physical layer.

2.4 Routing and Spectrum Allocation

2.4.1 Spectrum continuity and contiguity constraints

Establishing a lightpath in an Elastic Optical Network (EON) requires solving the complex Routing and Spectrum Allocation (RSA) problem. In traditional fixed-grid DWDM networks, resource allocation—known as Routing and Wavelength Assignment (RWA)—was primarily concerned with satisfying a simple wavelength continuity requirement [13]. However, the evolution to flexible-grid EONs introduces highly restrictive dual constraints that transform resource allocation into an NP-hard computational challenge [13]. Regardless of the specific allocation policy utilized, every valid assignment in an EON must fundamentally satisfy three primary physical constraints [2]:

- **Spectrum Continuity:** This constraint dictates that the exact same frequency spectrum (the specific index of the allocated spectrum slots) must be utilized and remain identical across every single link along the end-to-end routing path [14]. However, in multifiber infrastructures featuring parallel links, this constraint presents a crucial operational nuance [3]. While the frequency index must remain identical end-to-end, the optical signal is permitted to shift spatially between parallel physical fibers at each hop, provided there are no internal switching restrictions at the optical cross-connects [3].

- **Spectrum Contiguity:** The frequency slots assigned to a single lightpath must be strictly adjacent to one another within the frequency domain to form a contiguous "Slot Block". In modern implementations, algorithms treat individual frequency slots as fixed at a minimum width (typically 12.5 GHz) and group them contiguously to precisely match the variable bandwidth demands of the traffic [2].
- **Spectrum Non-Overlapping:** To prevent signal interference and data corruption, this constraint ensures that any specific frequency slot on a given physical fiber is allocated to a maximum of one active lightpath at a time [3].

The tight coupling of the continuity and contiguity constraints often leads to severe spectrum fragmentation. Because of the strict contiguity requirement, having enough total available spectrum on a path is insufficient if those slots are scattered and fragmented. To enforce these rules without blocking new requests, advanced RSA algorithms manage both constraints simultaneously by treating the network state as a unified "allocation matrix". This allows the controller to search for the first available block of contiguous slots that spans uniformly across all physical edges along the chosen multi-hop route [15].

2.4.2 Existing RSA algorithms

As established, the strict enforcement of spectrum continuity and contiguity transforms network resource allocation into a highly restrictive, NP-hard computational challenge[13]. To prevent severe spectrum fragmentation and the unfair blocking of high-capacity requests, intense research has been dedicated to developing sophisticated Routing and Spectrum Allocation (RSA) algorithms[16]. Over time, these algorithms have evolved from basic wavelength assignment models into complex, multi-metric optimization strategies designed to maximize network throughput and accommodate diverse, dynamic traffic demands[2]. Early approaches sought to decouple the routing and assignment phases to mitigate computational bottlenecks. Foundational algorithms employed greedy heuristics and K-shortest paths to maximize single-hop traffic while strictly enforcing standard wavelength continuity[4]. As networks transitioned to flex-grid EONs, researchers introduced offline traffic sorting heuristics to groom traffic more efficiently. For instance, the Minimum Hops with Least Slots (MHLS) algorithm prioritizes connection requests by calculating a sorting parameter[13]:

$$NP = Hops \times Slots$$

By serving "smaller" requests first, the algorithm minimizes overall blocked traffic[13]. More dynamic approaches have integrated custom cost functions to balance multiple network metrics simultaneously. For example, the `FRSO-RSA` algorithm evaluates candidate alternative routes by actively steering traffic away from congested or unreliable paths using the following cost function[17]:

$$cost_c = \alpha \times PF_c + (1 - \alpha)OU_c$$

where PF_c represents the physical failure rate of the path, OU_c denotes the spectrum occupancy rate, and α acts as an adjustable weight parameter[17]. To combat the specific issue of contiguity-induced fragmentation, recent literature has shifted toward "next-state-aware" algorithms. Metrics such as Maximum Spectrum Completeness (MSC) are used to evaluate how a current assignment will impact future capacity, actively selecting routes that preserve large, unbroken blocks of spectrum [18]. Other strategies introduce highly dynamic link-weighting mechanisms that actively draw traffic toward links harboring the largest unfragmented spectrum slices rather than just looking at total available capacity [19]. When fragmentation inevitably occurs during the network's lifecycle, complex defragmentation algorithms utilizing Integer Linear Programming (ILP) and Resource Dependency Digraphs (RDD) are deployed to mathematically recalculate and shift existing lightpaths without altering their physical routes[20]. Furthermore, advanced meta-heuristics like Genetic Algorithms and probabilistic models like Markov Decision Processes (MDP) have been explored to navigate massive solution spaces and predict network states[21], [22].

2.5 Problems with Parallel Links in RSA

2.5.1 Problem statement

While the Routing and Spectrum Allocation (RSA) problem in optical networks has been extensively studied for years, the comprehensive integration of multi-fiber or parallel links between adjacent nodes remains an underexplored area. Technologies such as link aggregation and bandwidth duplexing have become increasingly common, introducing multiple physical connections between devices to meet escalating capacity demands. Specifically, to aggregate a higher number of optical channels, multiple parallel connections are routinely established between a muxponder and a Reconfigurable Optical Add-Drop Multiplexer (ROADM). A critical physical constraint in this architecture is that the same spectrum cannot be reused across parallel links that terminate at the same degree of a ROADM device without causing severe interference. Traditional RSA algorithms primarily focus on path computation, spectrum continuity, and spectrum contiguity across single-fiber logical links. However, they frequently fail to account for the spatial complexity introduced by multi-fiber or parallel-link topologies. Current RSA algorithms already face significant scalability and efficiency drawbacks in complex, highly meshed topologies; introducing parallel links exacerbates these challenges dramatically. In a multi-graph network featuring a uniform or variable number of parallel links between adjacent devices, the total number of possible path combinations from a source to a destination increases exponentially. Because signals routed over the same ROADM degree cannot share frequencies, a network controller must exhaustively calculate and verify the availability of every frequency slot across every parallel edge. Computing paths in such multi-dimensional scenarios can easily overwhelm current algorithms, resulting in massive computational overhead, unacceptable latency, and ultimately, a failure in dynamic lightpath establishment. Consequently, there remains a critical research gap in developing routing strategies that can navigate the spatial complexity of parallel links without crippling network efficiency.

2.6 Current approaches and limitation

To address the NP-hard complexities of optical network management, theoretical research has developed sophisticated Routing and Spectrum Allocation (RSA) algorithms designed to mitigate severe spectrum fragmentation and maximize throughput. These strategies have evolved from foundational, decoupled wavelength assignment models and offline traffic sorting heuristics, into dynamic, multi-metric optimization approaches that actively steer traffic away from congested or unreliable paths. Recent literature also heavily emphasizes "next-state-aware" metrics, dynamic link-weighting, and complex defragmentation models. However, extensive taxonomic reviews reveal a prevailing structural limitation: the vast majority of these theoretical solutions are designed exclusively for one-dimensional, single-fiber-per-link topologies. When applied to multi-fiber scenarios, these models must treat parallel links either as physically invalid aggregated bandwidth or as entirely separate logical links, inevitably triggering a state-space explosion and unmanageable computational complexity.

In practical deployments, industry-standard SDN controllers like ONOS[23] and OpenDaylight (ODL)[24] attempt to manage these topologies using robust, graph-based routing engines, yet they fundamentally struggle with the same spatial constraints. In both controllers, optical connections are treated as strictly matched port-to-port links; thus, multiple parallel fibers between the same devices are mapped as entirely distinct and independent topological edges. When computing routes via standard K-Shortest Path algorithms, the topology engine outputs every permutation of these parallel links as a completely independent path sequence. Consequently, the subsequent spectrum allocation phase applies strict isolation, evaluating spectrum continuity and contiguity sequentially on a strict per-path basis. If an allocation attempt experiences wavelength contention on one parallel link, the controller discards the effort and must iteratively re-run the entire RSA verification from scratch on the next topological path permutation. In complex, highly meshed topologies featuring multiple parallel links, this exhaustive, sequential path-by-path iteration causes exponential computational overhead, often leading to unacceptable latency and dynamic provisioning timeouts. These systemic shortcomings in both theoretical models and practical SDN implementations highlight a critical research gap, presenting the opportunity to design a deterministic, spatially-aware parallel optical controller within the TeraFlowSDN ecosystem that evaluates parallel ports concurrently to bypass sequential iteration and enable highly scalable lightpath establishment.

2.7 Related work

A substantial body of literature exists on optimizing RSA in Elastic Optical Networks (EONs). Previous research has extensively modeled multi-hop routing constraints, fragmentation minimization, and modulation format adaptation. However, literature addressing the specific algorithmic handling of parallel links within a centralized, vendor-agnostic SDN architecture remains sparse. Most existing works assume simple graph topologies (one link between two nodes) or rely on simplifying assumptions that violate ITU-T flex-grid parameters. Consequently, there is a distinct necessity for a novel, stateful approach—such as a dedicated bitmap-based parallel controller—to bridge this gap between theoretical RSA and practical, scalable deployment in modern multi-link infrastructures.

3 Problem Definition and Proposed Algorithm

This chapter formalizes the Routing and Spectrum Allocation (RSA) challenge within multi-graph optical topologies and presents the core algorithmic solutions developed for this thesis. It details the methodologies used to model physical networks with parallel links, compute required bandwidth and slots, discover optimal and alternative paths, and dynamically allocate spectrum using a novel bitmap intersection approach.

3.1 Problem Overview

In modern optical networks, computing lightpaths between transponders and ROADMs requires resolving the Routing and Spectrum Allocation (RSA) problem. While standard algorithms can find paths across simple topologies, significant computational bottlenecks arise when topologies feature multiple parallel links between devices. Because optical channels transmitted to the same degree of a ROADM cannot reuse identical frequencies without conflict, the controller must exhaustively calculate and track the exact spectrum slot availability across all parallel permutations. The proposed solution addresses this by introducing a parallel optical controller that utilizes advanced graph modeling and bitwise spectrum calculations to efficiently manage these parallel paths.

3.2 Formal Problem Statement

TBD (Note: I have to define the mathematical formulation to represent the problem here, such as defining the network as a graph G=(V,E), defining the sets of available frequencies, and writing the objective function to minimize spectrum fragmentation or blocking probability.)

3.3 Graph Modelling with Parallel Links (topologies)

To enable efficient path computation and expansion, the physical network topology retrieved from the TeraFlowSDN database is modeled locally into a multi-directed graph using the NetworkX Python library. Only active optical devices are considered for this graph-building phase. Because NetworkX does not feature a default device-port-link formation, a custom mapping system was developed to link devices and endpoints. The controller generates multiple graph variations to serve different computational needs:

- `G_FREE`: A multi-directed graph that strictly includes "free" links where the optical channel status is OCH==USED[false]. This graph allows multiple parallel links between the same nodes.
- `G_simple_free`: A simplified version of the graph that filters out parallel links to find unique, distinct paths from the source to the destination.
- `G_simple`: The raw graph used to determine all possible topological paths without filtering for link utilization, which is later expanded to discover parallel combinations.

During the generation of these graphs, the controller fetches all links and creates a device-endpoint map based on TeraFlowSDN database records. For each endpoint, the channel information is retrieved to provide the algorithm with complete details regarding supported frequencies.

3.4 Proposed RSA Algorithm

3.4.1 Bandwidth & Slot Computation

Before pathing can occur, the controller must determine the exact amount of spectrum required for the requested optical intent. The bitrate for this operation is considered in GHz. The required bandwidth, B_r, is calculated using the following formula:

$$B_r = \frac{R_b}{n}(1 + \alpha)$$

where $n = log_2(M)$, and M represents the specific modulation format utilized. While several factors influence the required bandwidth, this standard formula simplifies the process for the controller. Once the bandwidth is known, the algorithm determines the number of required slots. Adhering to the minimum measurement unit defined by the ITU-T G.694.1 standard, the slot granularity is fixed at 6.25 GHz. The number of required slots, N, is calculated as:

$$N = \frac{B_r}{6.25}$$

3.4.2 Path Discovery

The path discovery phase utilizes the constructed NetworkX graphs to find viable routes. First, Dijkstra's algorithm is executed on the `G_FREE` graph to guarantee that the primary shortest path relies exclusively on unused optical links.

To discover alternative path combinations, a dedicated `find_paths()` function generates the `G_simple_free` graph, which includes all links, and executes the `all_simple_paths()` algorithm to find all possible distinct paths between the source and destination. Subsequently, an `expand_path()` function takes these lists of nodes and systematically computes all possible parallel link combinations between every two nodes along the route.

3.4.3 Bitmap-Based Spectrum Allocation

To manage the high-resolution ITU-T flex-grid granularity efficiently, the spectrum status is represented mathematically using a bitmap approach. Based on the detected optical band (e.g., C-Band, L-Band) and the corresponding minimum and maximum frequencies of the channels, the maximum bitmap value is generated using the formula:

$$bitmap - value = 2^N - 1$$

Because this produces an extremely large integer that cannot be natively stored as a numerical data type in standard relational databases like PostgreSQL, it is stored as a string and processed as an integer purely within the Python backend.

The RSA computation is handled by the `rsa_bitmap_pre_compute()` function. It processes the designated path and optical links, calculating the overall minimum and maximum frequency boundaries to generate a reference bitmap. The algorithm then performs a bitwise intersection across each consecutive hop in the path against the reference bitmap. This logical intersection rapidly identifies continuous blocks of available slots. Slots are categorized into status flags: AVAILABLE for free slots, `IN_USE` for allocated but non-restrictive fat channels (like OMS), FULL for consumed OCH endpoints, and UNAVAILABLE for slots outside the device's hardware capabilities.

3.4.4 Slot Acquisition & Assignment

Following the successful computation of an available spectrum block, the algorithm proceeds to the acquisition phase. It traverses each hop in the selected path and checks the current bitmap value stored in the `perform_rsa` cache. The algorithm then removes the alignment, converts the binary data back into an integer, and initiates a database update. The assignment is finalized by securely writing the updated state to the database via the context-client.

3.5 Complexity Analysis

`TBD` (Note: I have to write the time and space complexity of your algorithm here. For example, the Big-O notation for the networkX shortest path calculation on your multi-graph, and the complexity of the bitwise intersection across H hops and S slots).

4 System Design and Implementation

This chapter provides a detailed technical overview of the system architecture and the specific software implementations developed to support parallel optical links. It outlines the extensive modifications made to the TeraFlowSDN ecosystem, the creation of a standalone RSA computation module, and the deployment of the novel ParallelOpticalController microservice.

4.1 Overall System Architecture

The proposed solution is built entirely within the TeraFlowSDN (TFS) microservices ecosystem. TFS operates as a cloud-native, Kubernetes-based SDN controller, making it highly modular and extensible. To solve the Routing and Spectrum Allocation (RSA) problem for parallel links, the system architecture was augmented with a newly developed ParallelOpticalController service. This service interfaces heavily with the existing ContextService (for retrieving topology and endpoint database states) and the DeviceService (for communicating with the physical and emulated optical hardware). The architecture ensures that lightpath requests are intercepted by the ParallelOpticalController, which computes the optimal routing and spectrum allocation before instructing the DeviceService to push XML configurations to the underlying transponders and ROADMs.

4.2 Modifications to TeraFlowSDN

4.2.1 Device Service Enhancements (OCDriver, OpenROADMDriver)

To perform accurate device-based spectrum filtration, the controller requires detailed hardware telemetry. The existing OCDriver (for OpenConfig devices) was heavily modified to extract device information directly from the device's XML configuration files rather than relying solely on static JSON descriptors.

1. A new device-specific filter mechanism was introduced, establishing a device_filters package containing classes like DefaultDevice.py and CNITTestROADMDevice.py.

2. The OCDriver.py was updated to execute a two-phase extraction process: it first retrieves the model_name from the device, and then maps this model to the appropriate filter class to extract transponder and port values accurately.

3. Similar enhancements were made to the OpenROADMDriver, filtering out internal mapping ports and correctly marking client ports as OCH transport types and line ports as OMS transport types.

4.2.2 Context Service & Database Schema Extensions

Because the original database lacked fields for manufacturer and hardware specifications, significant schema extensions were introduced to ContextService.

1. The DeviceModel.py table was updated to include mandatory fields: mfg_name, model, and serial_no, alongside nullable fields for hardware_version and software_version.

2. The EndPointModel.py was extended to include an index string and a transport_type. The index acts as a unique identifier (e.g., "TP1_11") generated by combining a sanitized device name and the port index, resolving previous endpoint mapping ambiguities.

3. The OpticalChannelModel was upgraded to store crucial RSA metrics: `min_frequency`, `max_frequency`, `flex_slots`, and `bitmap_value`. These schema changes were formally defined in the context.proto protocol buffers and propagated throughout the gRPC services.

4.3 RSA Computation Module (RSATools, ITUStandards)

To centralize the mathematical and logical rules of optical networking, two primary Python helper modules were developed: ITUStandards.py and RSATools.py.

- **ITUStandards.py:** This module codifies the strict ITU-T G.694.1 standards. It defines constants such as the 193.1 THz anchor frequency and the 6.25 GHz slot granularity. It also encapsulates enumerations for frequency ranges across all optical bands (O, E, S, C, L) and slot availability statuses (AVAILABLE, UNAVAILABLE, `IN_USE`).

- **RSATools.py:** Functioning as the core logic engine, this module handles the dynamic RSA calculations. It includes functions like detect_band() to identify the smallest optical band containing a given frequency, and compute_rsa_params() to determine the minimum and maximum frequencies and required flex slots. Crucially, it calculates the base hardware spectrum bitmap using the formula $B_{\text{map}} = 2^{\text{slots}} - 1$ and dynamically generates spectrum slot tables for both routing computations and WebUI visualization.

4.4 Parallel Optical Controller Microservice

4.4.1 Service Architecture & API Design

The ParallelOpticalController was engineered as an autonomous microservice within TFS. It is containerized using Docker and deployed into the Kubernetes cluster alongside existing services. Following the deployment, both the serviceservice and webuiservice undergo rollout restarts to discover the new controller's environment variables and navigation endpoints.

4.4.2 Graph Construction with NetworkX

When an optical intent is received, the controller does not rely on static routing tables; instead, it dynamically reconstructs the network graph. Fetching endpoint and optical link data via the Context client, it utilizes the networkX package to build complex multi-directed graphs. The module `topology.py` is responsible for defining graph nodes (active optical devices) and generating distinct graph models, such as `G_FREE` for links where OCH is unused, and `G_simple` for raw path discovery.

4.5 Integration with Existing TFS Services

The controller interfaces seamlessly with the ContextService to read the current network state. During the "Slot Acquisition" phase, the controller updates the flow parameters in the database via the Context client, ensuring the allocated bitwise spectrum blocks are officially locked and recorded.

4.6 Topology & Link Management (Parallel Link Support)

A major implementation hurdle was supporting topologies with parallel links natively. The traditional method of mapping links via `endpoint_uuid` often failed when deploying topologies dynamically due to hashing mismatches. To resolve this, link descriptors were upgraded to support an `endpoint_index` fallback mechanism. Furthermore, to maintain a clean graphical user interface without overlapping link lines, parallel links are grouped and visually represented using an integer multiplier format, such as xN, detailing the exact number of parallel connections existing between two adjacent devices.

4.7 WebUI Enhancements (Spectrum Visualization)

To expose the advanced capabilities of the backend, the TFS Web User Interface underwent significant improvements.

- The device details template (detail.html) was modified to include the newly extracted Index and Transport Type metrics.
- A dedicated "Endpoint Frequency View" template was engineered, featuring a comprehensive "Frequency Capabilities Card" that displays wavelength ranges, bandwidth, and the specific optical band name.
- Most notably, the WebUI now features a real-time, scrollable "Spectrum Allocation Table". This table displays every individual 6.25 GHz slot within the device's band, utilizing color-coded badges to indicate slot status: green for AVAILABLE, yellow for `IN_USE`, gray for UNAVAILABLE (outside hardware bounds), and blue for the standard ANCHOR frequency.

5 Testbed Setup and Deployment

This chapter outlines the practical environment established to validate the proposed Routing and Spectrum Allocation (RSA) algorithms and the ParallelOpticalController. It details the configuration of the physical servers, the initialization of the Docker networking layer, the setup of emulated optical devices via XML data models, and the final microservices deployment procedure utilizing Kubernetes (MicroK8s).

5.1 Physical/Virtual Infrastructure

The experimental testbed relies on a distributed server architecture hosted at the University of Pisa (UniPi). Two primary servers, designated as Monster and Mascara, are utilized to balance the computational load between the SDN control plane and the emulated data plane.

To facilitate communication between the TeraFlowSDN controllers and the emulated devices, an isolated Docker bridge network is established. The creation of this network on the Monster server requires specifying a dedicated subnet and IP range:

```
# On secondary server
sudo docker network create --driver=bridge \
--ip-range=10.100.0.0/16 \
--subnet=10.100.0.0/16 \
-o "com.docker.network.bridge.name=brTFS" netbrTFS
```

Listing 5.1: Creating docker network

To ensure the Mascara server can route traffic to this newly created Docker subnet, static IP routing is configured. The following commands define the route through the respective gateway IP:

```
# Executed on Primary Server
sudo ip route add 10.100.0.0/16 via 131.114.54.73
route -n # Verification of the routing table
```

Listing 5.2: Creating route to the docker network

Furthermore, due to operating system constraints on Ubuntu 22.04 regarding network persistency, explicit iptables rules must be inserted on the Monster server to accept incoming traffic from the Mascara node:

```
# Executed on UniPi Server: Monster
sudo iptables -I DOCKER-USER -s 131.114.54.72 -d 10.100.0.0/16 -j ACCEPT
sudo iptables --list # Verification of firewall rules
```

Listing 5.3: Adding exceptions for teraflow packets

[This section will be written properly, because I have to introduce two setup. One for debian system with older versions of docker and the new one with ubuntu 22.04 and latest version of docker.]

5.2 Emulated Devices (Transponders, ROADMS)

Because deploying physical, large-scale flex-grid optical hardware is cost-prohibitive, the data plane is entirely simulated using advanced device emulators. These emulators accurately replicate the NETCONF/YANG interfaces of commercial optical equipment.

The configuration of these devices relies heavily on meticulously crafted XML files that define their hardware capabilities, physical ports, and operational frequencies.

- **Transponders (Muxponders):** The transponder configurations utilize the OpenConfig standard. For example, the `transponders_x4.xml` file defines a MellanoxSwitch device with multiple `QSFP56_DD_TYPE1` transceivers operating under the `ETH_400GBASE_ZR` standard. It explicitly details the optical channels, defining parameters such as the default 193.1875 THz frequency, target output power, and associated line ports.
- **ROADMs:** Reconfigurable Optical Add-Drop Multiplexers are configured to simulate specific optical degrees and frequency bands. The `roadm.xml` file dictates the input and output directional mapping of ports (e.g., ports 4101 and 5201 mapping to Degree D1). Crucially, it hardcodes the hardware bounds for the proposed RSA algorithm, explicitly defining the `MG_ON_PORT_MIN_FREQUENCY` as 193.0 THz and the `MG_ON_PORT_MAX_FREQUENCY` as 193.375 THz.

5.3 Network Topology Configuration (tfs descriptors)

To push the network topology into the TeraFlowSDN Context database, standardized JSON descriptors are utilized. These descriptors dictate the node identities, endpoint indices, and the optical links connecting them. To support the testing of topologies with multiple parallel links, the topology descriptor parsing mechanism was updated. Instead of relying strictly on unstable UUID hashing, the testbed utilises descriptors that reference the deterministic `endpoint_index` (e.g., `TP1_11`), allowing stable, repeatable topology generations with parallel edge definitions (e.g., xN multiplier configurations).

5.4 Deployment Procedure (TeraFlowSDN & ParallelOpticalController)

The entire TeraFlowSDN ecosystem, including the custom-built components, is deployed on a MicroK8s Kubernetes cluster. When deploying the novel ParallelOpticalController to a fresh TFS ecosystem, a strict five-step deployment pipeline is executed .

First, the Docker image for the new controller is built and pushed to the local container registry:

[**Deployment of TeraFlowSDN**]

6 Results and Evaluation

This chapter presents the performance evaluation and functional validation of the proposed Routing and Spectrum Allocation (RSA) framework. It assesses the operational success of the newly integrated ParallelOpticalController within the TeraFlowSDN (TFS) ecosystem and analyzes the efficiency of the proposed bitwise spectrum allocation algorithm across topologies featuring parallel optical links.

6.1 Evaluation Methodology & Test Scenarios

To rigorously evaluate the system, a series of test scenarios were executed within the established Docker-based emulated environment. The evaluation methodology focuses on injecting dynamic optical intents (lightpath requests) into the TeraFlowSDN controller and monitoring the system's response across multiple metrics.

The test scenarios utilize various network topologies, specifically focusing on multi-directed graphs where nodes (e.g., transponders and ROADMs) are connected via multiple parallel links, denoted by the xN multiplier format. The primary objectives of these tests are to validate the functional correctness of the device telemetry extraction, measure the computational efficiency of the path discovery algorithm, and statistically analyze the spectrum utilization under varying traffic loads.

TBD: (Detail the specific parameters of my test scenarios here, such as the exact number of nodes, the maximum number of parallel links per edge, the distribution of traffic requests, and the requested bandwidth/modulation formats used during the simulation.)

6.2 Functional Validation

Before analyzing quantitative performance, the functional correctness of the system was validated through backend logs and the enhanced Web User Interface (WebUI).

First, network connectivity and device emulation stability were verified by successfully pinging the emulated optical devices directly from the routing server (e.g., pinging 10.100.101.1 from the Mascara server). Following successful device discovery, the modified OCDriver and OpenROADM drivers successfully extracted the vendor-agnostic hardware parameters, which were confirmed via the Context Service database logs.

Furthermore, the functional validation is heavily supported by the newly developed visual representations in the TFS WebUI. When navigating to the device details page, the system successfully renders the "Endpoint Frequency View" template. This view dynamically displays the accurate Index and Transport Type of each port, alongside a comprehensive "Frequency Capabilities Card" outlining the specific optical band, wavelength range, and total flexible-grid slots.

Crucially, the "Spectrum Allocation Table" provides a real-time, color-coded visual validation of the RSA algorithm's output. The table accurately reflects the status of individual 6.25 GHz slots:

- Gray badges correctly identify UNAVAILABLE slots that fall outside the physical device's capabilities.
- Green badges represent AVAILABLE slots ready for provisioning.
- Yellow badges indicate `IN_USE` slots successfully allocated by the ParallelOpticalController.
- Blue badges accurately mark the ITU-T standard ANCHOR frequency (193.1 THz).

6.3 Computational Performance

TBD: (This section will present the execution time and computational overhead of the ParallelOpticalController. I must include graphs or tables showing the time taken to construct the NetworkX graph, the time taken to run Dijkstra's and the `all_simple_paths()` algorithms, and the speed of the bitwise spectrum intersection process as the number of parallel links increases.)

6.4 Spectrum Utilization Efficiency

TBD: (This section will evaluate how efficiently the flexible-grid spectrum is packed. I will discuss the fragmentation of the optical bands and provide metrics on how the proposed algorithm maximizes the use of 6.25 GHz slots across complex topologies compared to traditional fixed-grid or sequential parallel-link checking methods.)

6.5 Throughput Analysis

TBD: (This section will analyze the network's overall throughput based on the blocking probability. I must present data comparing the volume of successfully allocated optical intents versus rejected requests due to spectrum continuity constraints. Include an analysis of network-level

efficiency by comparing the ratio of allocated slots to the total available slots across all active links.)

6.6 Comparison with Baseline/Existing Approaches

TBD: (Here, I will benchmark your ParallelOpticalController against the default, sequential RSA mechanisms found in standard SDN controllers (like ONOS, OpenDayLight or the baseline TeraFlowSDN OpticalController). Provide comparative charts highlighting improvements in path computation time, reduction in blocking probability, and overall algorithmic efficiency when handling the xN parallel link scenarios.)

6.7 Statistical Analysis

TBD: (This section requires a deeper mathematical and statistical evaluation of the results. I will provide an analysis of the blocking paths and theoretical channel capacity against the actual used capacity during peak simulation periods.)

6.8 Discussion of Results

TBD: (I will provide a comprehensive summary and interpretation of the findings from the previous sections. Discuss whether the research objectives were met, highlight the most significant improvements brought by our parallel RSA algorithm, and acknowledge any unexpected behaviors or minor anomalies observed during the testing phase.)

7 Conclusion and Future Work

This concluding chapter summarizes the overarching achievements and technical contributions presented throughout this thesis. It critically reflects on the practical constraints and structural limitations encountered during the implementation phase and outlines potential avenues for future research to further refine the management of flexible-grid optical networks within TeraFlowSDN.

7.1 Summary of Contributions

As the demand for high-capacity telecommunications infrastructure continues to grow, optimizing the allocation of optical spectrum across complex topologies has become paramount. This thesis directly addressed the computational bottlenecks associated with Routing and Spectrum Allocation (RSA) over parallel optical links—a critical area often overlooked or inefficiently handled by traditional Software-Defined Networking (SDN) controllers.

The primary contribution of this research was the successful design, development, and deployment of the ParallelOpticalController, a stateful microservice fully integrated into the TeraFlowSDN (TFS) ecosystem. By leveraging the NetworkX library to construct multi-directed graphs, the controller natively supports and efficiently routes optical intents across topologies featuring multiple parallel edges. To resolve the heavy computational burden of high-resolution spectrum mapping, the system introduced a novel bitmap-based intersection algorithm capable of rapidly computing the availability of contiguous 6.25 GHz slots across multiple hops.

To ground this algorithmic advancement in practical hardware realities, this work significantly enhanced the TeraFlowSDN DeviceService and ContextService. The OCDriver (OpenConfig) and OpenROADM drivers were upgraded with custom model-based filters to extract vendor-agnostic telemetric data—such as operating frequencies, transport types (OCH, OMS), and hardware specifications—directly from XML configurations.

Finally, these backend capabilities were made accessible through a comprehensive upgrade to the TeraFlowSDN WebUI. The introduction of the dynamic "Spectrum Allocation Table" provides network administrators with real-time, color-coded visual feedback regarding the status (e.g., AVAILABLE, `IN_USE`, UNAVAILABLE) of individual ITU-T G.694.1 slots, drastically improving overall network transparency and operational user experience.

7.2 Limitations

TBD: More sophisticated device-based filtration, stateful characteristics of the controller, comparatively slower DB operations with intense topological intents ,generating a reply with the results and update the device with a NETCONF request.

7.3 Future Work Directions

TBD

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